CBSE Class XII Board Examination 2027 — Expected Paper

PHYSICS (Theory) · Code 042

Time allowed: 3 hours **Maximum marks: 70**

General instructions

1. This paper has 33 questions in five sections, A–E. All questions are compulsory.
2. Section A: Q1–16 are MCQs of 1 mark each (Q13–16 are Assertion–Reason). Section B: Q17–21, 2 marks each. Section C: Q22–28, 3 marks each.
3. Section D: Q29–30 are case-based questions of 4 marks each. Section E: Q31–33, 5 marks each.
4. This is an original practice paper built on the latest CBSE sample-paper pattern. It is not an official CBSE paper.

Section A — Multiple choice (1 mark each). Questions 13–16 are Assertion–Reason.

1. Two electric field lines never intersect each other because **[1]**

(a) the field is zero at the point of intersection
(b) the field would have two directions at the same point
(c) like charges repel
(d) field lines are imaginary

2. The equipotential surfaces of a point charge are **[1]**

(a) planes
(b) concentric spheres
(c) coaxial cylinders
(d) ellipsoids

3. A wire of resistance 4 Ω is bent into a circle. The resistance between two diametrically opposite points is **[1]**

(a) 4 Ω
(b) 2 Ω
(c) 1 Ω
(d) 0.5 Ω

4. When the temperature of a metallic conductor increases, the relaxation time of free electrons **[1]**

(a) increases
(b) decreases
(c) remains unchanged
(d) becomes zero

5. A charged particle enters a uniform magnetic field perpendicular to it. Its path is **[1]**

(a) a straight line
(b) a parabola
(c) a circle
(d) a helix

6. A material with a small negative magnetic susceptibility is **[1]**

(a) paramagnetic
(b) diamagnetic
(c) ferromagnetic
(d) ferrimagnetic

7. Lenz's law is a consequence of the conservation of **[1]**

(a) charge
(b) momentum
(c) energy
(d) mass

8. In a purely inductive AC circuit, the current **[1]**

(a) is in phase with the voltage
(b) leads the voltage by π/2
(c) lags the voltage by π/2
(d) lags the voltage by π

9. For an electromagnetic wave in vacuum, the ratio E_0/B_0 equals **[1]**

(a) c
(b) 1/c
(c) c^2
(d) √c

10. The power of a lens of focal length −25 cm is **[1]**

(a) +4 D
(b) −4 D
(c) −0.25 D
(d) +0.25 D

11. In Young's double-slit experiment, if one slit is closed, the screen shows **[1]**

(a) uniform illumination
(b) the diffraction pattern of a single slit
(c) fringes of double width
(d) no light at all

12. The maximum kinetic energy of photoelectrons depends on the **[1]**

(a) intensity of the incident light
(b) frequency of the incident light
(c) time of exposure
(d) area of the metal surface

13. **Assertion (A):** The electric potential everywhere inside a charged hollow conducting sphere is constant. **[1]**
Reason (R): The electric field inside the sphere is zero.

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** Bohr's model cannot explain the spectra of multi-electron atoms. **[1]**
Reason (R): Bohr's model ignores electron–electron interactions.

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** The resistivity of a semiconductor decreases with increase in temperature. **[1]**
Reason (R): The number of charge carriers increases with temperature.

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

16. **Assertion (A):** A convex lens always forms a real image. **[1]**
Reason (R): A convex lens converges a parallel beam of light.

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

Section B — Very short answer (2 marks each)

17. State Gauss's law in electrostatics. A charge q is placed at the centre of a cube. Find the electric flux through (i) the whole cube and (ii) one face. **[2]**

18. Resistors of 4 Ω and 12 Ω are connected in parallel across a 6 V battery of negligible internal resistance. Find (i) the current drawn from the battery and (ii) the power dissipated in the 4 Ω resistor. **[2]**

19. Define mutual inductance. Write its SI unit and two factors on which the mutual inductance of a pair of coils depends. **[2]**

20. A ray of light passes through an equilateral glass prism of refractive index $\sqrt{2}$ at minimum deviation. Find the angle of minimum deviation. **[2]**

21. An electron and a proton have the same de Broglie wavelength. Which of them has more kinetic energy? Justify. **[2]**

Section C — Short answer (3 marks each)

22. Derive an expression for the electric potential at a point on the axial line of an electric dipole. How does it vary with distance r for $r \gg a$? **[3]**

23. State Kirchhoff's two rules. Use them to obtain the balance condition of a Wheatstone bridge. **[3]**

24. Using the Biot–Savart law, derive the magnetic field on the axis of a circular current loop of radius R at distance x from its centre. Hence write the field at the centre. **[3]**

25. A series LCR circuit with L = 2.0 H, C = 32 μF and R = 10 Ω is connected to a 230 V variable-frequency AC source. Find (i) the resonant angular frequency, (ii) the rms current at resonance and (iii) the Q-factor. **[3]**

26. Draw a labelled ray diagram of a compound microscope with the final image at the near point. Write the expression for its magnifying power. **[3]**

27. Write Einstein's photoelectric equation. Draw the graph of stopping potential against frequency and state what its slope and intercept represent. **[3]**

28. Draw the binding energy per nucleon curve. Use it to explain the release of energy in nuclear fission and fusion. **[3]**

Section D — Case-based questions (4 marks each)

29. **Case study: p-n junction diode.** When a p-n junction forms, diffusion of majority carriers creates a depletion region and a barrier potential. Forward bias lowers the barrier and a large current flows; reverse bias raises it and only a tiny current flows. Diodes are used as rectifiers. **[4]**

(i) What happens to the width of the depletion region in forward bias?

(ii) What is the typical barrier potential of a silicon diode?

(iii) Which charge carriers cause the reverse current?

(iv) A full-wave rectifier is fed with 50 Hz AC. What is the frequency of the ripple in the output?

30. **Case study: total internal reflection.** When light travels from a denser to a rarer medium at an angle of incidence greater than the critical angle, it is totally reflected. This is used in optical fibres, prisms and the sparkle of diamonds. **[4]**

(i) State the two conditions for total internal reflection.

(ii) Find the critical angle for a medium of refractive index $\sqrt{2}$.

(iii) On what principle does an optical fibre work?

(iv) Why does a diamond sparkle?

Section E — Long answer (5 marks each)

31. (a) Derive the capacitance of a parallel-plate capacitor with plate area A and separation d when a dielectric slab of dielectric constant K and thickness t ($t < d$) is placed between the plates. **[5]**

(b) A charged capacitor is disconnected from the battery and a dielectric slab (K) fills the space between its plates. How do the capacitance, potential difference and stored energy change?

32. (a) Explain the principle and working of a moving coil galvanometer. Why is a radial magnetic field used? **[5]**

(b) A galvanometer of resistance 50 Ω gives full-scale deflection for 2 mA. How will you convert it into a voltmeter of range 0–10 V?

33. (a) Derive the expression for fringe width in Young's double-slit experiment. **[5]**

(b) In a YDSE, $\lambda = 600$ nm, slit separation 0.5 mm and screen distance 1 m. Find the fringe width. What happens to it if the apparatus is immersed in water ($n = 4/3$)?

Marking scheme — Physics (042)

Value points and mark split

Section A — Multiple choice (1 mark each). Questions 13–16 are Assertion–Reason.

1. (b) At an intersection the field would have two directions, which is impossible.

2. (b) Concentric spheres centred on the charge.

3. (c) Two 2 Ω halves in parallel give 1 Ω.

4. (b) More frequent collisions with vibrating ions shorten τ, so resistivity rises.

5. (c) qvB provides the centripetal force: r = mv/qB.

6. (b) Diamagnetic (e.g. bismuth, copper).

7. (c) Energy: the induced current opposes the change that produces it.

8. (c) It lags the voltage by π/2.

9. (a) $E_0/B_0 = c$.

10. (b) P = 100/f(cm) = −4 D (diverging lens).

11. (b) Interference disappears; only single-slit diffraction remains.

12. (b) $K_{max} = h\nu - \varphi_0$.

13. (a) E = −dV/dr = 0 inside, so V is constant and equal to its surface value.

14. (a) Electron–electron repulsion is significant in multi-electron atoms.

15. (a) More electron–hole pairs form at higher T, which outweighs the shorter relaxation time.

16. (d) An object inside the focus gives a virtual image (magnifier), so A is false.

Section B — Very short answer (2 marks each)

17. Total flux through a closed surface = (charge enclosed)/ε_0. (i) $\Phi = q/\varepsilon_0$ (ii) by symmetry, each face gets $q/6\varepsilon_0$. [1 + ½ + ½]

18. $R_p = (4\times12)/16 = 3\ \Omega$; $I = 6/3 = 2$ A. Current in 4 Ω = 6/4 = 1.5 A; $P = V^2/R = 36/4 = 9$ W. [1 + 1]

19. $M = \Phi_2/I_1$: flux linked with one coil per unit current in the other. Unit: henry (H). Depends on the number of turns, area and separation/orientation of the coils, and the permeability of the core. [1 + ½ + ½]

20. $n = \sin[(A + \delta m)/2] / \sin(A/2) \Rightarrow \sqrt{2} \times \sin30° = \sin[(60° + \delta m)/2] \Rightarrow \sin[(60° + \delta m)/2] = 1/\sqrt{2} \Rightarrow (60° + \delta m)/2 = 45° \Rightarrow \delta m = 30°$. [2]

21. $\lambda = h/p$, so equal λ means equal momentum. $K = p^2/2m$, so the lighter electron has more kinetic energy ($K_e/K_p = m_p/m_e \approx 1836$). [1 + 1]

Section C — Short answer (3 marks each)

22. Charges $-q$ at $-a$ and $+q$ at $+a$; point P at distance r on the axis. $V = (1/4\pi\varepsilon_0)[q/(r - a) - q/(r + a)] = (1/4\pi\varepsilon_0)\cdot 2qa/(r^2 - a^2) = p/[4\pi\varepsilon_0(r^2 - a^2)]$. For $r \gg a$, $V = p/(4\pi\varepsilon_0 r^2)$, so $V \propto 1/r^2$ (compared with $1/r$ for a point charge). [2 + 1]

23. Junction rule: $\Sigma I = 0$ at a junction (charge conservation). Loop rule: $\Sigma(IR) = \Sigma E$ around a closed loop (energy conservation). Balanced bridge ($I_g = 0$): loop ABDA gives $I_1P = I_2R$ and loop BCDB gives $I_1Q = I_2S$. Dividing, $P/Q = R/S$. [1 + 2]

24. $dB = (\mu_0/4\pi)\, I\, dl/(x^2 + R^2)$. Perpendicular components cancel; axial components add: $dB \cos\theta$ with $\cos\theta = R/\sqrt{(x^2 + R^2)}$. $B = \mu_0 IR^2/[2(x^2 + R^2)\text{^}\{3/2\}]$ along the axis. At the centre ($x = 0$), $B = \mu_0 I/2R$. [2½ + ½]

25. (i) $\omega_0 = 1/\sqrt{(LC)} = 1/\sqrt{(2 \times 32 \times 10^{-6})} = 125$ rad s^{-1}. (ii) At resonance $Z = R$, so $I_{rms} = 230/10 = 23$ A. (iii) $Q = \omega_0 L/R = (125 \times 2)/10 = 25$. [1 + 1 + 1]

26. Diagram: objective forms a real, inverted, magnified image A′B′ inside the focus of the eyepiece; the eyepiece forms the final virtual image at D. $m = m_o \times m_e = (v_o/u_o)(1 + D/f_e)$ in magnitude; for a short tube length, $m \approx (L/f_o)(D/f_e)$. [2 + 1]

27. $h\nu = \varphi_0 + K_{max} = h\nu_0 + eV_0$. $V_0 = (h/e)\nu - \varphi_0/e$: a straight line. Slope = h/e (the same for all metals); intercept on the ν-axis = threshold frequency ν_0; intercept on the V-axis = $-\varphi_0/e$. [1 + 1 + 1]

28. The curve rises steeply for light nuclei, peaks at about 8.75 MeV ($A \approx 56$, Fe), then falls slowly to about 7.6 MeV for $A \approx 238$. Fission: a heavy nucleus ($A \approx 236$) splits into middle-mass nuclei with higher BE/A, releasing about 200 MeV. Fusion: light nuclei combine into a nucleus with higher BE/A, releasing energy. [1 + 1 + 1]

Section D — Case-based questions (4 marks each)

29. (i) It decreases. (ii) About 0.7 V. (iii) Minority carriers. (iv) 100 Hz. [1 × 4]

30. (i) Light must go from a denser to a rarer medium, and $i > i_c$. (ii) $\sin i_c = 1/\sqrt{2} \Rightarrow i_c = 45°$. (iii) Repeated total internal reflection at the core–cladding boundary. (iv) Diamond has a very small critical angle (about 24.4°), so light entering it is totally reflected many times before leaving. [1 × 4]

Section E — Long answer (5 marks each)

31. (a) $E_0 = \sigma/\varepsilon_0$ in air, $E = E_0/K$ in the slab. $V = E_0(d - t) + (E_0/K)t = (Q/A\varepsilon_0)(d - t + t/K)$. $C = Q/V = \varepsilon_0 A/(d - t + t/K)$. [3] (b) Q is constant: C becomes KC, V becomes V/K, and $U = Q^2/2C$ becomes U/K (energy decreases). [2]

32. (a) A current-carrying coil in a magnetic field feels a torque $\tau = NIAB$. A spring gives a restoring torque $k\varphi$, so at equilibrium $\varphi = (NAB/k)I$ and the deflection is proportional to current. A radial field (from concave poles and a soft-iron core) keeps the plane of the coil parallel to B, so the scale is linear. [3] (b) $R = V/I_g - G = 10/0.002 - 50 = 4950\ \Omega$, connected in series. [2]

33. (a) Path difference at height y on the screen: $\Delta = yd/D$. Bright fringe: $yd/D = n\lambda \Rightarrow y_n = n\lambda D/d$. Fringe width $\beta = y_{n+1} - y_n = \lambda D/d$. [3] (b) $\beta = (600 \times 10^{-9} \times 1)/(0.5 \times 10^{-3}) = 1.2$ mm. In water $\lambda' = \lambda/n$, so $\beta' = \beta/n = 0.9$ mm. [2]